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## **Degradable Materials Comprehensive Testing Provides Valuable Insights for Multistage Fracturing Completion System Applications in HPHT Wells**

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### **Abstract**

Degradable technology has been used in Multi-Stage Fracturing (MSF) operations since early 2000. The efficiency gained from utilization and availability of a wide variety of degradable materials has accelerated its adoption in the oil and gas industry. MSF applications, such as ball drop MSF frac sleeves or plug and perf stimulation are the most prevalent practices where degradable technologies are utilized. The understanding of the degradation mechanism and factors affecting the same are paramount in material selection and operational planning. With real-well scenarios and extensive lab testing, this paper aims to comprehend the challenges in the utilization of dissolvable frac balls and proposes adequate solutions.

In the context of this paper, terms "degradable" and "dissolvable" are used interchangeably, as is normal in the industry. And the focus will be on dissolvable metallic alloy frac balls which is magnesium based, although the behavior of other metal-based alloys is fairly similar. Fundamentally, all degradable products operate on a similar mechanism – a chemical reaction with water which produces a microscale powder as a byproduct.

Ball drop frac sleeves utilized in MSF completion systems have been a proven method to improve operational efficiency. The utilization of dissolvable balls has simplified operations by reducing wireline or Coiled Tubing (CT) interventions. Nevertheless, in deep gas wells, MSF system components have to stand for higher differential pressures and temperatures. Dissolution mechanisms largely depend on the downhole temperature, amount of water in contact with the dissolvable ball, and the fluid salinity.

The results underscored the importance of analyzing well parameters and fluid properties accurately to obtain predictable behavior from dissolvable frac balls utilized in MSF completion systems. The key observations related to medium, fluid contact, dissolution mechanisms, engagement of frac ball with ball seat will be elaborated, which can be implemented for improving fracturing operational performance in High Pressure High Temperature (HPHT) wells. These observations were then correlated with lab tests in representative fluids and temperatures, which served to confirm the hypotheses. In addition, the sealing properties of these balls along with their interface with ball seat have also been explored with a view to understanding an optimum method to be employed in operations.

Dissolvable technologies for MSF, while extremely efficient and useful to simplify and accelerate frac operations, is not an exact science. The benefits from utilization of this technology, however, far outweigh the potential uncertainties. The uncertainties can further be reduced by understanding the environment of operation and planning for viability and dissolution times accordingly. Further, availability of contingency measures and forward planning will greatly enable the advantages of utilizing a completion system focused on degradable material performance.

## Introduction

The implementation of degradable materials for frac plugs and frac balls in hydraulic fracturing process has become a widespread practice, aiming for enhancing efficiency. Dissolvable materials include polymer-based compounds and metal alloys demonstrates high pressure resistance and temperature tolerance, while ongoing development focuses on improving degradation consistency and mechanical performance. The diversity in material types, operating conditions, and performance characteristics suggests that different degradable materials may be suited for various well environments and applications. For MSF applications, several degradable solutions including metal alloys, hybrid materials, thermoset composites, acid-resistant materials, degradable elastomers, and fully degradable component are currently utilized.

Field studies report successful implementation of degradable frac plug and frac ball technologies in a HPHT tight gas formations. As per [Al-Obaid et al., 2023](#), the frac plug maintained structural integrity during proppant fracturing and degraded naturally post-operation, eliminating the need for costly milling interventions. It was demonstrated the frac plug ability to hold pressure for 32 hours under extreme conditions (285°F and exceeding 12,500 differential pressure), validating its effectiveness in reducing operational risks and complexities. In another field study, acid-resistant dissolvable frac balls sustained pressures up to 8,000 psi at temperatures reaching 300°F in multi stage acid stimulation treatment ([Arias et al., 2022](#)).

Laboratory tests detailed degradable polymers such as polyurethane elastomers—with tensile strengths above 5,000 psi, over 500% elongation, and high compression modulus greater than 13,000 psi ([Burdzy, 2021](#)); thermoset polymer composites exhibiting compressive strengths around 37k psi and tensile strengths near 14k psi ([Kovalchuk et al., 2022](#); [Zhao et al., 2023](#)). [Aviles et al. \(2013\)](#) addressed the challenge of frac ball failure in MSF operations by introducing degradable alloy balls. These balls, designed to disintegrate uniformly in well fluids through an electrochemical process, eliminate the risk of stuck balls and costly mill-out operations. The material's predictable degradation ensures reliable stage isolation during stimulation and optimal well performance post-job. This innovation enhances the efficiency and productivity of multistage stimulation operations, particularly in tight formations. Metal-based systems include degradable alloys that disintegrate via electrochemical action. And dissolvable magnesium or aluminum alloys that offer tunable degradation rates and reliable sealing ([Takahashi et al., 2018](#); [Tu et al., 2019](#)). Advances are evident in the formulation of hybrid systems that combine polymers and alloys to achieve high pressure resistance, in the development of materials with controlled degradation, and in laboratory demonstrations that balance strength with degradability. Challenges center on achieving consistent degradation under extreme downhole conditions, balancing acid resistance with mechanical performance, and ensuring compatibility with existing equipment. These studies collectively delineate the materials, performance metrics, and operational parameters for degradable frac plugs and frac balls in MSF and MSF completion systems.

Previous work quoted above highlights the advancements in degradable materials and designs, offering solutions to persistent challenges in hydraulic fracturing and MSF completion systems. These innovations improve operational efficiency, reduce costs, and enable access to complex formations, contributing to the sustainability and productivity of MSF. Common conclusions from these studies are material degradation

behaviors are specific to degradation mechanism such as chemical dissolution, corrosion, hydrolysis, etc. Key factors influencing degradation can be listed as;

- Temperature range
- Pressure conditions
- Fluid media and composition (type of fracturing fluids, salinity, pH)
- Material composition and chemical additives
- Flowback interference

Performance characteristics key performance metrics are;

- Mechanical properties; compressive strength, tensile strength, elasticity, elongation, and compression modulus
- Sealing performance; holding time under certain pressure
- Degradation time&rates and simulations
- Comparative performance against traditional materials
- Validation methods; laboratory tests and field operations

Despite the growing interest in degradable materials, there is still a need for further research to fully understand their behavior and performance under various downhole conditions. The existing literature highlights the potential benefits of degradable balls, but more studies are needed to address the concerns about their reliability and effectiveness. Our current work elaborates on further development by comprehensive lab testing and field trial of degradable frac balls utilized in a novel MSF completion system that is detailed in the next section.

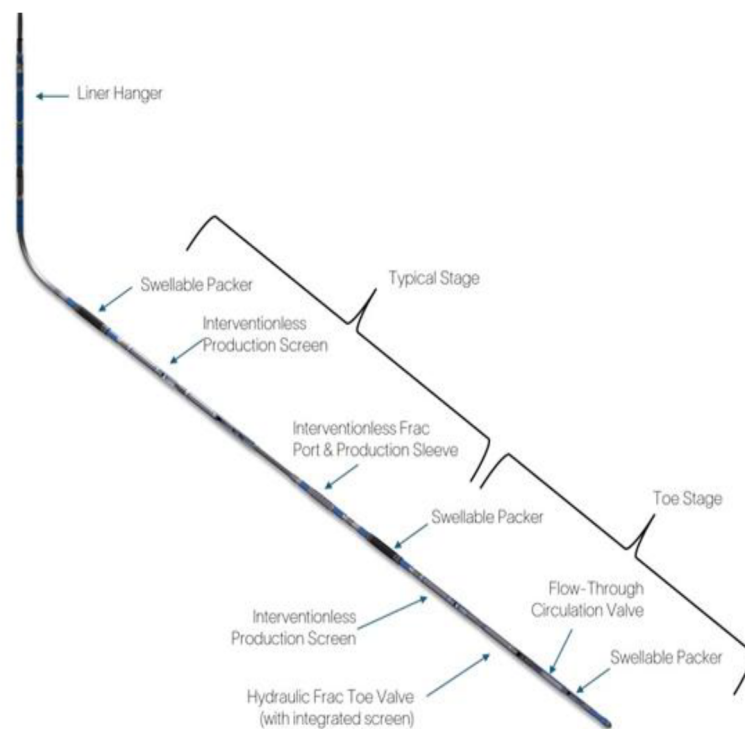
## Completion Introduction

The MSF with Sand Control (MSF w/SC) completion system, as elaborated by [Rodoplu et al. in 2023](#), is an advanced MSF completion solution designed for achieving fracturing and sand control at the same time in formations particularly those with low permeability and high sanding tendencies. This system is a permanent part of the lower completion, set across zones of interest to facilitate efficient fracturing while controlling sand production.

Key features of the system include the use of swell packers to isolate compartments in the open hole, ensuring precise control over fracturing zones. Each stage is equipped with re-closable frac ports and screened sleeves, which allow for controlled fluid injection and sand filtration. The first stage utilizes a hydraulic frac port, while subsequent stages utilizes ball-activated frac sleeves, enabling sequential opening without well intervention. The system's flexibility allows for customization, with the ability to configure stages with multiple screened sleeves to suit specific well requirements. The system's design also includes components like float shoes & collars to prevent backflow, and toe circulation sub that has a role in setting the liner hanger and packer. The lower completion uses a high pressure rated liner 4-1/2", sized for a 5-7/8" open hole, while the upper completion features a monobore design for operational simplicity. This configuration ensures continuous fracturing and effective sand control, making it an optimal solution for uncemented openhole vertical and horizontal wells.

Further insights of the system's components and functionality were discussed by [Rodoplu et al. in 2019](#), which provides a comprehensive understanding of the system's design and operational capabilities. In this initial version, post-fracturing operations involve closing frac ports and opening screened sleeves to control

sand and proppant during production. This process necessitates CT operations to mill the frac balls and seats first, and then shift close frac sleeves and shift open screened sleeves ensuring smooth well operation. Recently this system has undergone an enhancement as seen in Fig. 1 and was designed to be interventionless due to operational needs in which degradable frac balls were the main component making the system completely intervention free.



**Figure 1—Enhanced MSF with sand control completion system schematic**

In this new configuration, each frac sleeve consists of multiple sliding sleeves and ball seats. The first stage is hydraulically activated as it is in other open hole MSF completion systems. The toe valve is activated with absolute downhole pressure and is activated by increasing the pressure above a certain threshold in the tubing string.

The first sleeve in the 2<sup>nd</sup> stage will shift downward when activated with the first ball to allow the frac treatment to be performed. The second sleeve will also shift downward to close the frac ports and open the screened ports, and it is also activated remotely with a second ball. The third sleeve can be used at a later stage as a remedial solution in case of early water or gas breakthrough, as it will enable to close the production port. This new solution brings significant advantages compared to previous version on the completion system.

In addition to the frac sleeves, a series of ball activated on/off screen sections can be used to increase the screen coverage in each stage as seen in Fig. 2. To ensure no additional intervention is required, the system is designed with dissolvable balls to allow immediate production without adding any additional restriction compared to the string. And to ease the deployment before the treatment, it is designed to be rotatable while running in hole.



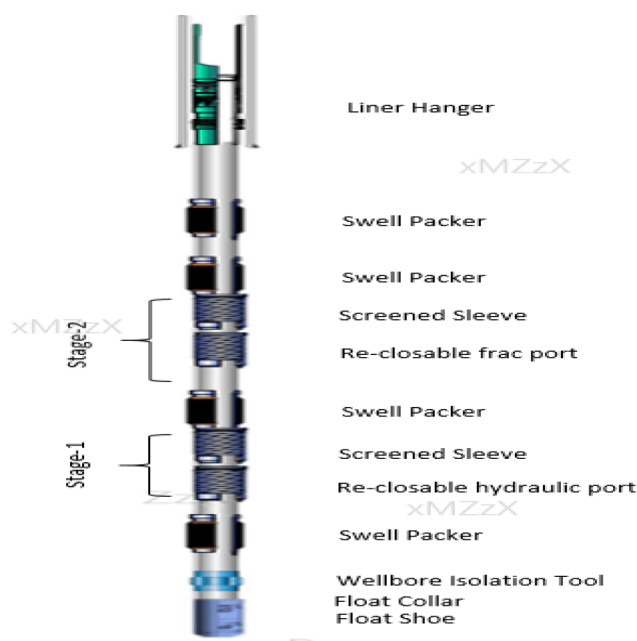


Figure 2—MSF with sand control completion system schematic designed for 2 stages in a vertical well

The high-level operational sequence is provided below for reference;

1. Run the completion to depth
2. Open the toe valve as per the design and perform the first treatment
3. Drop the first dissolvable ball and confirm the first stage is closed and the first production port is opened
4. Optionally drop a 2<sup>nd</sup> dissolvable ball to open the next screen in the 1<sup>st</sup> stage if additional screen is placed in the stage
5. Drop the larger dissolvable frac ball, open the frac port and perform the treatment in the 2<sup>nd</sup> stage
6. Drop the next dissolvable ball, confirm the frac port is closed and the production port is opened
7. Repeat if there is another screen in the stage
8. Repeat 6, 7 & 8 as needed
9. Flow back the well

The main challenge was finding a suitable dissolvable frac ball that is compatible with the MSF w/SC completion system and performing seamlessly in HPHT downhole conditions. Next section focuses on the methodology and lab testing followed to find the proper solution.

## Methodology and Lab Testing

The first step during the development phase was to look for a degradable material that could provide an acceptable level of performance given the challenging well conditions such as temperatures over 300 °F and differential pressures in the excess of 13k psi. In addition, following were the main design specifications;

- Various ball sizes within the internal dimension requirements (min ID of the smaller seat >2.5" and max ball OD compatible with the casing drift 3.688")
- Strong mechanical properties with high sealing performance to sustain required differential pressures in the excess of 13,000 psi
- Dissolve within 2 days to be able to safely flow back the wells in KCl based fluid

Based on the completion configuration and actuation sequence, two types of balls are required for this application:

- Stimulation balls: These are the balls required for the stimulation operation. After landing, they need to sustain stimulation pressures of 13k-15k psi for 1-2 hours considering the duration of fracturing treatment
- Shifting balls: These are the balls that only perform a sleeve shifting function. They need to sustain 2k-3k psi for a short duration after landing

To ease the development, the same material was originally targeted for both operations. As the balls for stimulation and shifting had different pressure rating requirements, the overlap between the ball and the seat was modified to achieve two ball designs.

The material selection started with 6 different materials that were known to be potential candidates for the application: T, U, W, X, Y, and Z. The objective was to first identify materials that could dissolve slow enough in the KCl conditions, to retain integrity during the stimulation operations. Two coupons of each were tested in KCl for over 27 hours at maximum temperature. Only the X and Y coupons dissolved slowly enough for the operation. The U coupon did not dissolve at all, while the others were dissolving 50 times faster than the X coupon and 12 times faster than the Y coupon as seen in the [Fig.3](#), [Fig.4](#), and [Fig.5](#). It is worth to note that there was still a ratio of 4 between the dissolution rate of the materials X and Y. The material X was chosen as a base case for this system due to its slower dissolution rate.



Figure 3—Coupons for material selection (example with material Y) – before test



Figure 4—Coupons for material selection (example with material Y) – 1 hr soaking



Figure 5—Coupons for material selection (example with material Y) – 24 hr soaking

The second step was to confirm that the seat geometry and the ball material were able to achieve the maximum differential rating required for each ball type. This helped identifying the best geometry and overlap to achieve the objectives.

The qualification tests were then performed on each stimulation ball in various sizes (3.379, 3.036, and 2.725 inches diameter) following this protocol; heat the setup up to 200 °F for 30-60 min, hold 15k psi differential for 30 min and then hold 12k psi for 90 min. This was deemed close enough to a standard fracturing stimulation operation between the higher fracture initiation pressure and the cooling effect at the ball depth, when pumping at a high rate. All these tests were meeting the initial specifications. The balls were slightly deformed as expected as seen in Fig. 6 due to the load generated by the high pressure, but could all come off seat easily after the tests.



Figure 6—An example of a X dissolvable ball after the mechanical tests

For the shifting balls, a less rigorous protocol was followed, given the required operational criteria: heat the setup up to 200 °F for 30-60 min, and hold 3k psi differential for 10 min. As it was possible to correlate the amount of stress with the various overlaps, only one ball was tested. This proved to be successful as well.

Lastly, a non-dissolvable ball (aluminum) was tested as a comparison. The test was also successful following the same stimulation ball protocol, but the pressure to dislodge the ball from the seat was excessive, and therefore not considered as a favorable design.

To confirm that there would not be any unexpected issue at a higher temperature, the same dissolvable X balls were also tested at 15k psi/300 °F for one hour as seen in Fig.7 below. These tests were promising,

but more pressure was required to dislodge the balls. As the balls are dissolvable, this was not deemed an issue compared to the aluminum, since a partially dissolvable ball can easily come out of the seat.



Figure 7—Dissolvable frac ball (a) Before 300 °F test; (b) After 300 °F test

All tests were done with 2% KCl, however it was later identified that, in some instances, the characteristics of the stimulation fluid was much closer to fresh water. Hence, a new material was needed to ensure that the balls would eventually dissolve. The Y material was a good choice at the time, as its dissolution rate in fresh water is about the same as the dissolution rate of the X material in KCl. It also displays slightly better mechanical properties.

The previous tests performed on the X balls were repeated with the Y ball:

- Stimulation balls: heat the setup up to 200 °F for 30-60 min, hold 15k psi differential for 30 min and then hold 12k psi for 90 min.
- Stimulation balls: Test a new Y ball for 1 hour at 15k psi/300 °F to confirm viability at higher temperature
- Shifting balls: heat the setup up to 200 °F for 30-60 min, hold 3k psi differential for 10 min

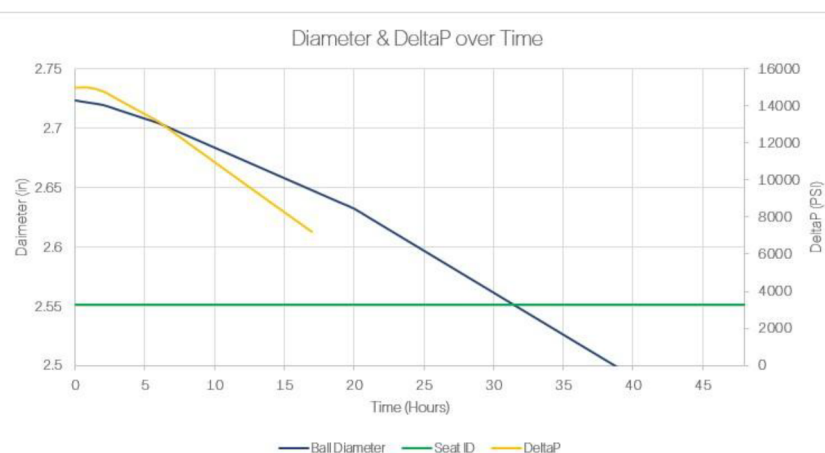
These tests were also successful, and the pressure required to dislodge the balls was even lower than with the X balls.

During the planning of the operations, the dissolution tests were repeated on both X and Y coupons using the stimulation fluid mixed with water from an offset water well in the same field. These tests were done at expected downhole temperature of 300 °F.

The dissolution rate for the Y material was in line with the previous testing and considered ideal, but the X ball did not dissolve in this fluid. These tests confirmed the suitability of the Y balls for both the stimulation and the shifting operations. Based on additional data gathered during the various mechanical tests performed in conjunction with all the dissolution tests, a simulated performance envelope was produced, mainly for the stimulation ball. This simulation as seen in Fig. 8 relied on a uniform dissolution of the balls in the fluid at downhole temperature.

During the frac operation, there is surface work required after dropping the ball such as rigging up wellhead isolation tool before performing the frac, it may take up to 12 hours in some cases. In this worst-case scenario, there were concerns about the ball integrity after this extended time downhole. The suitability of the Y balls for the stimulation operation was therefore reviewed.





**Figure 8—Dissolution simulation for the Y stimulation ball (Fresh water-based fluid, balls dropped at ambient, 1 hour before reaching the seat at around 300 °F, uniform degradation)**

As the dissolution rates for the Y balls in fresh water and the X balls in KCl were equivalent, and as the frac operation was the most critical, it was decided to use the X balls during the stimulation to avoid any potential integrity issue. To accelerate the dissolution of the X ball after the frac operation, the stimulation program added KCl in the flush.

For the shifting balls, there was no issue with delays on surface, so the Y balls were selected as their dissolution rate was acceptable in fresh water-based fluid.

In summary, a single degradable material did not meet all the required operational criteria. Adequate planning with the right options and testing, enabled the selection of the ideal degradable material for each step of the application.

## Deployment of the System and Job Execution

The novel MSF w/SC completion system was run successfully in a trial well as a 2 stages completion with the rig without any issues. After waiting for the swell packers swelling, proppant fracturing treatments were conducted riglessly to stimulate the formation with a high sanding tendency. Treatment was designed using around 250K lbs of 20/40 size high strength proppant (without any resin coated proppant) up to 6 ppa concentration and 45 bpm pumping rate. Based on pressure signatures during deployment and treatment execution, data acquired during fracturing & miscellaneous pumping operations, wireline/slickline interventions, and opening/closing of frac sleeves, production screened sleeves, the completion system was evaluated for its performance.

## Post-Trial Analysis

After the various slickline runs, it was confirmed that the balls had not dissolved even after several days in the well at high temperature. This was unexpected and contrary to the expectations; therefore several deliberations were held between various stakeholders to ascertain the likely cause.

To understand the root cause of the issue, the following steps were taken:

1. Manufacturing and batch quality were checked, and the balls were found to be as per the required material specifications. Another ball from the same batch was exposed to a dissolution test and the dissolution process was seen to start as low as 160 °F in the fresh water. This ruled out the possibility of an abnormal dissolution rate due to material anomalies.
2. Specific analysis of the formation properties was done in order to investigate other causes contributing to the issue. In a typical dissolution simulation, the results obtained on test coupons are extrapolated



to a sphere which generally has a large contact area with the dissolving fluid medium. In this case, it was observed that the formation was sub-hydrostatic, therefore the pressure above the ball (pushing it onto the seat) was higher than the pressure below the ball (formation pressure) as depicted in Fig. 9 below. Once the ports were opened, a restricted fluid flow path was preventing fluid circulation above the ball. Additionally, since the well was vertical, the ball could not easily roll off the seat like in a horizontal well case. Calculations showed that there was a differential pressure across the ball seat (above to below).

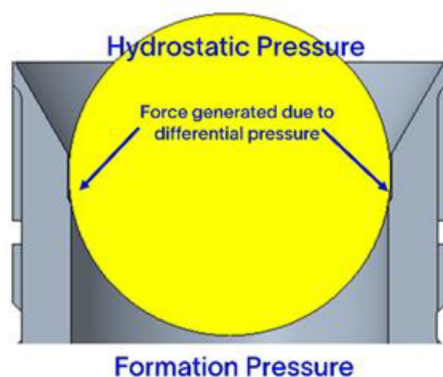


Figure 9—Depiction of dissolvable ball on the ball seat

The high likely scenario was quite different compared to the lab testing that had been performed. This is because lab testing assumes that the entire surface of the ball is in contact with the fluid, causing a fairly uniform dissolution over the surface area. In this real-well scenario, the largest diameter of the ball is pushed into the seat internal diameter as seen in the Fig.10. There is a limited fluid contact across this diameter. It is hypothesized that an unexpected result was created by this specific geometry;

- The limited fluid contact on the sides restricts the amount of material that will dissolve across the largest diameter.
- There is no chemical diffusion of fluid volumes above and below the seat.
- The by-products caused by ball degradation can accumulate in the small area above the seat limiting the dissolution rate further.

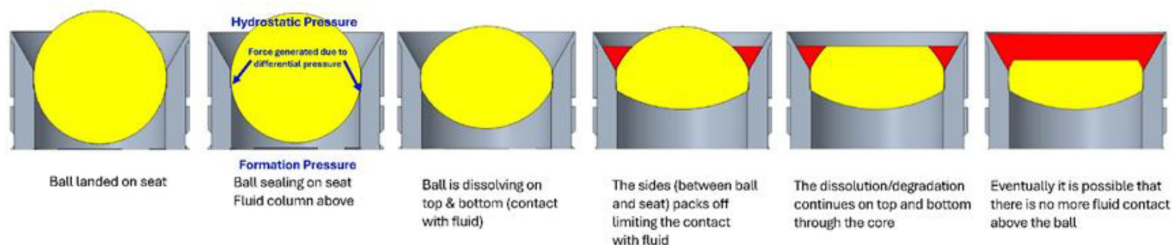


Figure 10—Non-uniform dissolution hypothesis showing how the ball is dissolved when sealing

As shown in the Fig. 10 below, this could cause the ball to dissolve only through the core and not over the entire surface area, which significantly extends the total dissolution period.

To prove this theory, additional tests were performed on the Y ball for confirmation. In the first set of tests, a ball was seated inside a fixture with KCl to accelerate the dissolution rate at 190 °F. To simulate the sub hydrostatic condition, the ball was pushed into the seat to create a seal. After 3.5 and 7.5 hours the ball was inspected to understand the dissolution mechanism.

After 3.5 hours, the ball was inspected but not dislodged from the seat. Debris from the dissolution reaction were seen accumulated around the ball sealing area as seen in Fig.11. This was in line with the hypothesis.



Figure 11—Approximately 3.6% mass loss after 3.5 hours

After 7.5 hours, more debris was seen accumulated on the seat and the ball was dislodged for inspection. The shape of the ball matched what was expected from the theory as seen in Fig. 12. The dissolution was seen to be non-uniform and least through the core.



Figure 12—Approximately 13.6% mass loss after 7.5 hours

More extended duration tests were performed to characterize the ball dissolution further in a real-well scenario. These tests were first performed on the shifting balls. For these tests, the original Y material was used. Nevertheless, in parallel, another potentially suitable material, Y2, was also tested. Both balls were setup in a fixture and tested as follows:

- Heat up the fixture to 300 °F while the ball was soaked in medium fluid
- Increase the differential pressure to 3,000 psi (above to below) and maintain for 5 min (time required to shift the sleeve)
- Maintain a differential pressure of 1,000 psi (above to below) at 300 °F, until the ball dissolved enough to allow fluid bypass.

Table-1 lists and summarizes the testing for the Y ball

Table 1—Y ball testing results

| Y Ball test step                                               | Time (in fluid) – min | Effective time at 300egF – min |
|----------------------------------------------------------------|-----------------------|--------------------------------|
| Heat Up                                                        | 118                   | 10                             |
| Hold (3,000 psi)                                               | 5                     | 5                              |
| Hold at Temperature (1,000 psi)                                | 281                   | 281                            |
| Extended test (temperature fluctuated due to a facility issue) | 2,756                 | 2,288                          |
| Total (minutes)                                                | 3,260                 | 2,684                          |
| Total (days)                                                   | 2.25                  | 1.86                           |

Before the test the ball diameter was 2.54" and its weight was 280.7 gr. After the test the biggest diameter was 2.499" and its weight was 190.7 gr (-32%) as seen in Fig. 13. Despite a significant mass loss, the ball could still not fall through the seat and the post-test shape confirmed a non-uniform degradation due to the seal at the seat.

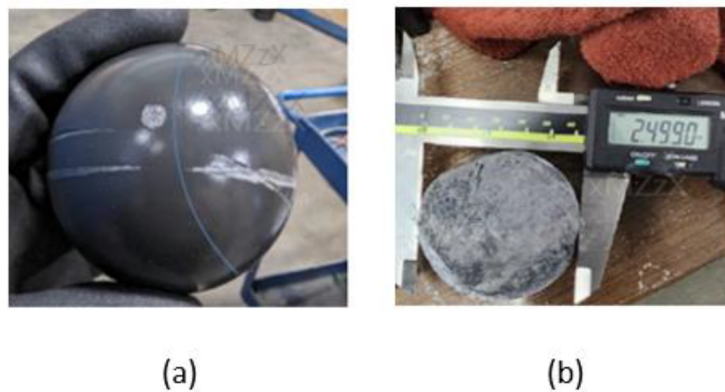


Figure 13—Dissolvable Y frac ball (a) Before the test; (b) After the test

The alternate material, Y2, behaved differently, as it is more suited to dissolve fast in fresh water. A ball of the same size was used in this test to have a good comparison. A similar protocol as for the Y material was followed, but the soaking time was increased to account for the worst-case scenario using a faster dissolving material.

Table-2 lists and summarizes the testing for the Y2 ball

Table 2—Y2 ball testing results

| Y2 Ball test step               | Time (in fluid) | Effective time at 300egF |
|---------------------------------|-----------------|--------------------------|
| Heat Up                         | 160             | 60                       |
| Hold (3,000 psi)                | 7               | 7                        |
| Hold at Temperature (1,000 psi) | 956             | 956                      |
| Extended test                   | 1,292           | 1,176                    |
| Total (minutes)                 | 2,550           | 2,334                    |
| Total (days)                    | 1.77            | 1.62                     |

The fixture was opened to inspect the ball shape after the total test time. The ball had fully dissolved as seen in Fig.14.



Figure 14—Full dissolution of Y2 ball after the test

For the stimulation X balls, the geometric dissolution behavior is similar and the dissolution rate for these balls is very low in fresh water, hence the earlier decision to add salinity in the form of KCl in the flush for fast dissolution after stimulation was still valid. In addition, the extended soaking period still represented a risk that needed to be properly addressed. This is why additional tests had to be completed to assess the compatibility of this material with the specific downhole conditions and the operational constraints.

## Summary of Findings & Remedial Actions

In vertical wells with sub-hydrostatic formation conditions, the ball dissolution mechanism is not the same as what is expected in normal pressured wells or horizontal wells, and this needs to be considered when designing a completion and a stimulation operation coming afterwards. Also, variations in HPHT environments and dissolution time will require extensive testing and evaluating different types of ball materials. In addition to this, various medium fluids should be tested to ensure proper isolation during stimulation, while also enabling the ball to dissolve as required.

Based on the results seen during the testing. The Y2 balls are chosen to be used for shifting operations on the next well, as these will dissolve quickly in fresh water. The Y balls will be used for stimulation operations. At the end of stimulation, KCl will be added in the flush fluid to increase the dissolution rate. This arrangement provides good isolation for an extended duration during the stimulation operations, while the fast dissolution post-stimulation also limits potential challenges during the flow back operations.

## Conclusion

This paper has provided a comprehensive review of the current state of knowledge on degradable materials specifically frac balls, highlighting their design, application, and performance. It has also identified areas for further research, such as material optimization, predictive modeling, and environmental impact studies, which are essential for advancing their use. The development of degradable materials faces diverse challenges, necessitating varied and innovative approaches. Main takeaways can be listed as follows;

- **Material Selection and Performance:** The choice of material significantly impacts the performance of dissolvable frac balls. Materials such as X and Y, with varying dissolution rates in different fluids, were tested. Material X showed slower dissolution in KCl based fluids, making it suitable for stimulation operations, while material Y performed better in fresh water, ideal for shifting operations. The introduction of material Y2, which dissolves rapidly in fresh water, offers a promising solution for stimulation operations.



- **Impact of Well Conditions:** The study revealed field conditions, particularly in vertical wells with sub-hydrostatic formation pressures, lead to non-uniform dissolution of frac balls. Limited fluid contact and differential pressure results in slower and uneven degradation, extending dissolution times beyond expectations and lab testing findings.
- **Adaptive Operational Strategies:** To ensure reliability, some strategies can be adopted from operational point of view such as adding KCl or similar to the flush fluid that accelerates the dissolution of frac balls post-stimulation, mitigating flowback challenges. Moreover, minimizing time of surface operations after dropping the dissolvable ball should be considered.
- **Importance of Testing:** Extensive lab testing and field trials are crucial for understanding material behavior under HPHT conditions. These tests are essential for validating material performance and ensuring compatibility with specific well environments.

The effective use of degradable balls in MSF completion systems should be based on careful analysis to ensure their suitability for specific applications, coupled with robust testing protocols and sufficient data to validate their performance. The field performance underscores the need for extensive testing and data collection to address these challenges comprehensively. By pursuing these points, the degradable material can further be developed for more efficient, reliable, and cost-effective solutions, ultimately enhancing the development of HPHT formations. Continued research and development are vital to overcoming the challenges associated with degradable materials and ensuring their successful application in the field operations.

## Nomenclature

|          |                                            |
|----------|--------------------------------------------|
| CT       | : Coiled Tubing                            |
| HPHT     | : High Pressure High Temperature           |
| ID       | : Inner Diameter                           |
| MSF      | : Multi Stage Fracturing                   |
| MSF w/SC | : Multi Stage Fracturing with Sand Control |
| OD       | : Outside Diameter                         |
| OH       | : Open Hole                                |

## References

- Al-Obaid, H., Al-Mulhim, B., Ashby, S., Alyousef, A. 2023. Degradable Plug Technology Holds Pressure for 32 Hours and Eliminates Risk of Intervention in Tight Gas Reservoirs. Paper presented at the SPE/IADC Middle East Drilling Technology Conference and Exhibition, Abu Dhabi, UAE, 23-25 May. SPE/IADC-214562-MS. <https://doi.org/10.2118/214562-MS>
- Arias, R., Bernal, R., Shakeel, M., Normand, M.. 2022. Successful Deployment of Acid-Resistant Dissolvable Frac Ball for Multi-Stage Acid Stimulation Application in an Offshore Multi-Lateral Well. Paper presented at the ADIPEC Conference and Exhibition, Abu Dhabi, UAE, 31 Oct.-3 Nov. SPE-211761-MS. <https://doi.org/10.2118/211761-MS>
- Aviles, I., Dardis, M., Jacob, G. 2015. Dissolvable Plug and Perf System Eliminates Mill-Outs in Multistage Stimulations. Journal of Petroleum Technology 1 June. SPE-0615-0038-JPT. *J Pet Technol* **67** (06): 38–40. <https://doi.org/10.2118/0615-0038-JPT>
- Burdzy, M. P. 2021. Aqueous Degradable Polyurethane Elastomers for Oil & Gas Applications. Paper presented at Offshore Technology Conference, Houston, TX, USA, 16-19 Aug. OTC-31186-MS. <https://doi.org/10.4043/31186-MS>
- Jin, Ningjing, and Qijun Zeng. "Dissolvable Tools in Multistage Stimulation." Paper presented at the SPE/IATMI Asia Pacific Oil & Gas Conference and Exhibition, Jakarta, Indonesia, October 17-19, 2017. SPE-186184-MS. <https://doi.org/10.2118/186184-MS>
- Raafat, Ahmed, Ahmed Samy, Ruggero Brambilla, and Carlos Monteiro. "Degradable Balls; A Successful Approach towards the Optimization of Multistage Completion Operations." Paper presented at the 14th Offshore Mediterranean Conference and Exhibition, Ravenna, Italy, March 27-29, 2019. OMC-2019-1198.

- Rodoplu, R. S., Al-Mohanna, K.S. 2019. Planning, Design and Optimization of CT Intervention for the First Successful Multistage Fracturing Completion with Sand Control System in the Region. Paper presented at the Abu Dhabi International Petroleum Exhibition & Conference, Abu Dhabi, UAE, 11-14 November. SPE-193114-MS. <https://doi.org/10.2118/197368-MS>
- Rodoplu, R. S., Brohi, G. I., Al-Mohanna, K.S., Qahtani, A. A. 2023. A New Multistage Fracturing Completion Approach for Sanding Formations: First Application in the Region. Paper presented at the International Petroleum Technology Conference, Bangkok, Thailand, 1-3 March. IPTC-22755-MS. <https://doi.org/10.2523/IPTC-22755-MS>
- Takahashi, Shinya, Aki Shitsukawa, and Masayuki Okura. "Degradation Study on Materials for Dissolvable Frac Plugs." Presented at the Unconventional Resources Technology Conference, Houston, Texas, July 23-25, 2018. URTEC-2901283-MS. <https://doi.org/10.15530/URTEC-2018-2901283>
- Tu, Huilin, Isaac Aviles, and Michael Dardis. "Degradable Materials for Multi-Stage Stimulation." Paper presented at the Offshore Technology Conference, Houston, Texas, USA, May 6-9, 2019. OTC-29282-MS. <https://doi.org/10.4043/29282-MS>
- Zhao, L, J. X. Ren, M. Yuan, Y. Liu, H. L. Liu, , C. J. Zhou, S. W. Wang, G. F. Ren, P. Cheng. 2023. Novel Hybrid Degradable Plugs to Enable Acid Fracturing at High Temperature High Pressure Conditions. Paper presented at the SPE Annual Technical Conference and Exhibition, San Antonio, Texas, USA, 16-18 October. SPE-214810-MS. <https://doi.org/10.2118/214810-MS>